

# Kingdom of Science: biology.pdf

## Penicillin Isolation

Penicillium fungi secrete beta-lactam metabolites inhibiting bacterial cell walls. Required: Moldy citrus/bread, dextrose nutrient broth, charcoal. Procedure: Inoculate sterile broth with Penicillium spores. Incubate 7 days. Filter broth through activated charcoal. Elute penicillin. Safety: Check under microscope.

## Ethanol Fermentation

Yeast ferments simple sugars into ethanol and CO<sub>2</sub>. Required: Mash, wild yeast, airlock. Procedure: Mash ripe fruits in water, pitch yeast, seal with water airlock for 10 days. Safety: Ensure CO<sub>2</sub> ventilation.